

Port No. Flat	Dual	Port symbol	Signal	Input/ Output	Voltage, etc.		Description
					Stop	Play	
59	3	Pφ1	M DATA	0	0	→	IC501 data } IC501 clock } LCV control and interpolation circuit mode IC501 load } setting
60	4	Pφ2	M CLK	0	5	→	
61	5	Pφ3	M LD	0	5	→	
62	6	PC2	MUTE	0	0	→	
63	7	P1φ	TR FWD	0	0	→	High during skip, etc.
64	8	P11	TR REV	0	0	→	Traverse forced forward signal high Traverse forced reverse signal high

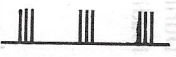
Port No.		Port symbol	Signal	Input/Output	Voltage, etc.		Description
Flat	Dual				Stop	Play	
19	27	P41	LCD CK	O	0		IC901 LCD driver clock
20	18	P42	DATA	O			LCD, LED data
21	29	P43		NC			
22	30	P5 ϕ	LED LOAD	O			LED data sending
23	31	P51	LED CK	O			IC7 LED decoder load
24	32	P52		NC			IC7 LED decoder clock
25	33	VDD		I	5V	→	Battery 5V
26	34	PC ϕ		O	5V	→	
27	35	PC1	CDPY OUT	O	5V	0	CD play signal, system control
28	36	P53		O	5V	→	Battery 5V detection
29	37	P6 ϕ		NC			
30	38	P61		NC			
31	39	P62		NC			
32	40	P63		NC			
33	41	P7 ϕ	<u>SENSE</u>	I			IC101 Optical servo state signal input
34	42	P71	<u>FLOCK</u>	I	5V	0	IC101 RF signal/no RF signal input; Low for RF signal input
35	43	P72	TEMP	I	5V	→	IC101 temperature sensor L: Stop command
36	44	P73	DTC (STAT)	I	0	0 (Pulse sometimes)	IC501 LCV state confirmation input
37	45	P8 ϕ	Low voltage detection	I	5V	→	IC2 (Reset circuit)
38	46	P81		NC			

Port No.	Port		Signal	Input/ Output	Voltage, etc.		Description
	Flat	Dual			Stop	Play	
39	47		P82	I		→	Traverse position confirmation
40	48		P83	I	0		
41	49		P9 ϕ	I			
42	50		P91	I	5V	→	
43	51		P92	I	5V	→	
44	52		P93	I	5V	→	
45	53		PA ϕ	O	5V	→	Loading solenoid control 0 when door starts opening 0 when door is open
46	54		PA1	O	5V	→	
47	55		PA2	NC			
48	56		PA3	O	5V	→	Power circuit control
49	57		PB ϕ	O	5V	0	Optical servo control signal
50	58		PB1	O	5V	0	
51	59		PB2	O	5V	0	
52	60		PB3	O	5	0	
53	61		OSC2			→	Crystal 4 MHz
54	62		OSC1			→	
55	63		X1	I	→		Same as <u>SENSE</u> input as to P7 ϕ
56	64		X ϕ	NC			
57	1		VSS	O	0	→	
58	2		P $\phi\phi$	NC			

TRE, BASS, CONT CN9 (6-pin) P3D only

Pin No.	Name of Pin	Stop	Play	Remarks
1	RNF	0	4.5V	When connected to DIN input
2	R $\ominus$	0	4.5V	
3	ROUT	0	4.5V	
4	LOUT	0	4.5V	
5	L $\ominus$	0	4.5V	
6	LNf	0	4.5V	

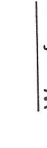
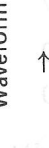
LCD Control Key Input CN10 (26-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	LCD LOAD	Waveform 	→	Digital 5V
2	LCD CLOCK		→	
3	DATA	5V	→	High when reset
4	5V	→	→	
5	NC	/	/	From system main amplifier illumination power
6	BLANK (LCD)	0	→	
7	GND	0	→	High pulse when key is ON
8	Illumination (LCD)	14.4V	→	
9	Q3	0	→	Voltage drops when LED lights.
10	Q2	0	→	
11	Q1	0	→	Repeats high and low during loading and eject.
12	Q ϕ	0	→	
13	INT	0	→	From system main amplifier illumination power
14	RM (Ramdom memory)	8V	→	
15	RAUSE	8V	→	Battery 5V
16	STOP	7V	8V	
17	PLAY	8V	7V	10V when reset button is ON
18	REPEAT	8V	→	
19	SCAN	8V	→	
20	DISC	0	→	
21	ALARM (NC)	/	/	
22	GND	0	→	
23	$\oplus$ 10V	10V	→	
24	Illumination	14.4V	→	
25	BUP 5V	5V	→	
26	RESET	0	→	

3-2 Microcomputer Pin Voltages (Stop, Play)

Microcomputer MN1554 CAC or MN1554

Sheet 1

Port No.		Port symbol	Signal	Input/Output	Voltage, etc.		Description
Flat	Dual				Stop	Play	
1	9	P12		O	5V	↑	During loading { H L } During eject { L H } Loading motor FWD/REV control
2	10	P13		O	5V	↑	
3	11	$\overline{\text{SYNC}}$		NC	5V	↑	Low key interrupt input when key input is entered
4	12	$\overline{\text{SIRO}}$		I	0		
5	13	$\overline{\text{IRO}}$	BLK CK	I	0		IC501 block clock (75 Hz)
6	14	$\overline{\text{SBT}}$	$\overline{\text{CLD CK}}$	I	5V		
7	15	$\overline{\text{SBO}}$		NC	5V		IC501 frame clock (7.53 kHz)
8	16	$\overline{\text{SBI}}$	$\overline{\text{SUB Q}}$	I	Waveform	Waveform	
9	17	$\overline{\text{RST}}$		I	5V	↑	IC501 subcode Q data
10	18	P2φ	CDPSIN	I	0	↑	
11	19	P21	HAR	I	0	↑	Toyota specifications H . . . Stop H . . . Stop (Traffic information)
12	20	P22	VAC+5V	I	5V	↑	
13	21	P23		NC			ACC power detection
14	22	P3φ	Qφ	I	0	↑	
15	23	P31	Q1	I	0	↑	Key input of 4-bit code
16	24	P32	Q2	I	0	↑	
17	25	P33	Q3	I	0	↑	IC901 LCD driver load LCD data sending
18	26	P4φ	LCD LOAD	O	0	↑	

Spindle Motor Input/Output CN3 (6-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	FAI (EC) output	1.5V	Waveform	
2	GND	0	→	
3	VCC (10V)	10V	→	
4	PC output	0		
5	FGP input	0	Waveform	
6	DCR	2.5V	→	

Focus Tracking Output CN4 (4-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	Tr ⊕	0	Waveform	 ±5VP-P
2	Tr ⊖	0	→	
3	Fo ⊕	0	Waveform	
4	Fo ⊖	0	→	

Optical Pickup Input/Output CN5 (11-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	G GND	0	→	
2	A2	0	→	
3	A3	0	→	
4	A1	0	→	
5	A4	0	→	
6	⊕ 5V	5V	→	Analog + 5V
7	RF	4V	→	Signal is so low in level that it is invisible.
8	GND	0	→	
9	PD	⊖ 5V	⊖ 3V	
10	LD	0	⊖ 3V	
11	⊖ 5V	⊖ 5V	→	

LPF Digital Power CN6 (3-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	GND	0	→	Digital 5V
2	⊖ 5V	⊖ 5V	→	
3	⊕ 5V	5V	→	

LPF Control Input/Output CN7 (9-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	PAM	0		De-emphasis disk 5V during play
2	GND	0	→	
3	LDG	Waveform	→	
4	RDG	Waveform	→	
5	D. EMPH	0	0	
6	⊕ 10V	10V	→	
7	⊖ 10V	⊖ 10V	→	
8	ROUT	0	Audio Waveform	
9	LOUT	0	Audio Waveform	

DC/DC Converter CN8 (6-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	V BAT	14.4V	→	Battery power
2	GND	0	→	
3	V ACC	14.4V	→	Accessory power 0 when disk is off
4	V CONT	5V	→	
5	⊖ 5V	– 5V	→	
6	⊖ 10V	– 10V	→	

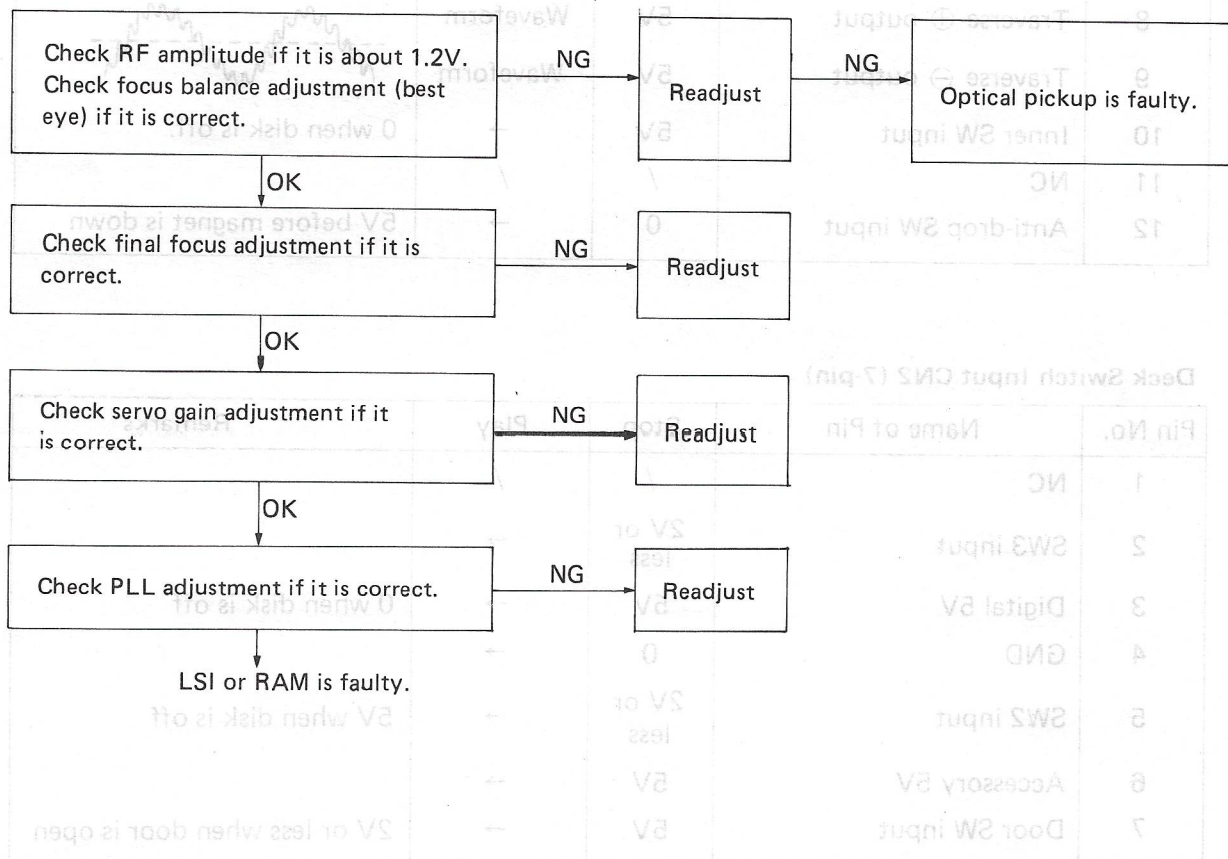
2-9 Readjustments

The set may be considered operating because the operation from reading the TOC to initiating play has been checked and found normal.

- Check the RF amplitude and eye pattern opening if they are sufficient.
- Check if offset adjustment is correct.
- Check if servo gain adjustment is correct.
- Check if the corrective capacity of the LSI is satisfactory.
- * For the adjustment procedures, refer to the adjustment part of the manual.

Check procedures and possible faulty points

Connect an oscilloscope as described in the adjustment procedures, and monitor the following signals.



3. APPENDIX

3-1 Connector Voltage (Stop, Play)

Deck Connector CN1 (12-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	Digital 5V	5V	→	0 when disk is off
2	Accessory 5V	5V	→	
3	GND	0	→	
4	Solenoid ⊕ output	14.4V	→	Battery 14.4V power
5	Solenoid ⊖ output	14.4V	→	0 when door is open
6	Loading motor ⊖ output	0	→	10V during eject
7	Loading motor ⊕ output	0	→	10V during loading
8	Traverse ⊕ output	5V	Waveform	
9	Traverse ⊖ output	5V	Waveform	
10	Inner SW input	5V	→	0 when disk is off.
11	NC	/	/	
12	Anti-drop SW input	0	→	5V before magnet is down

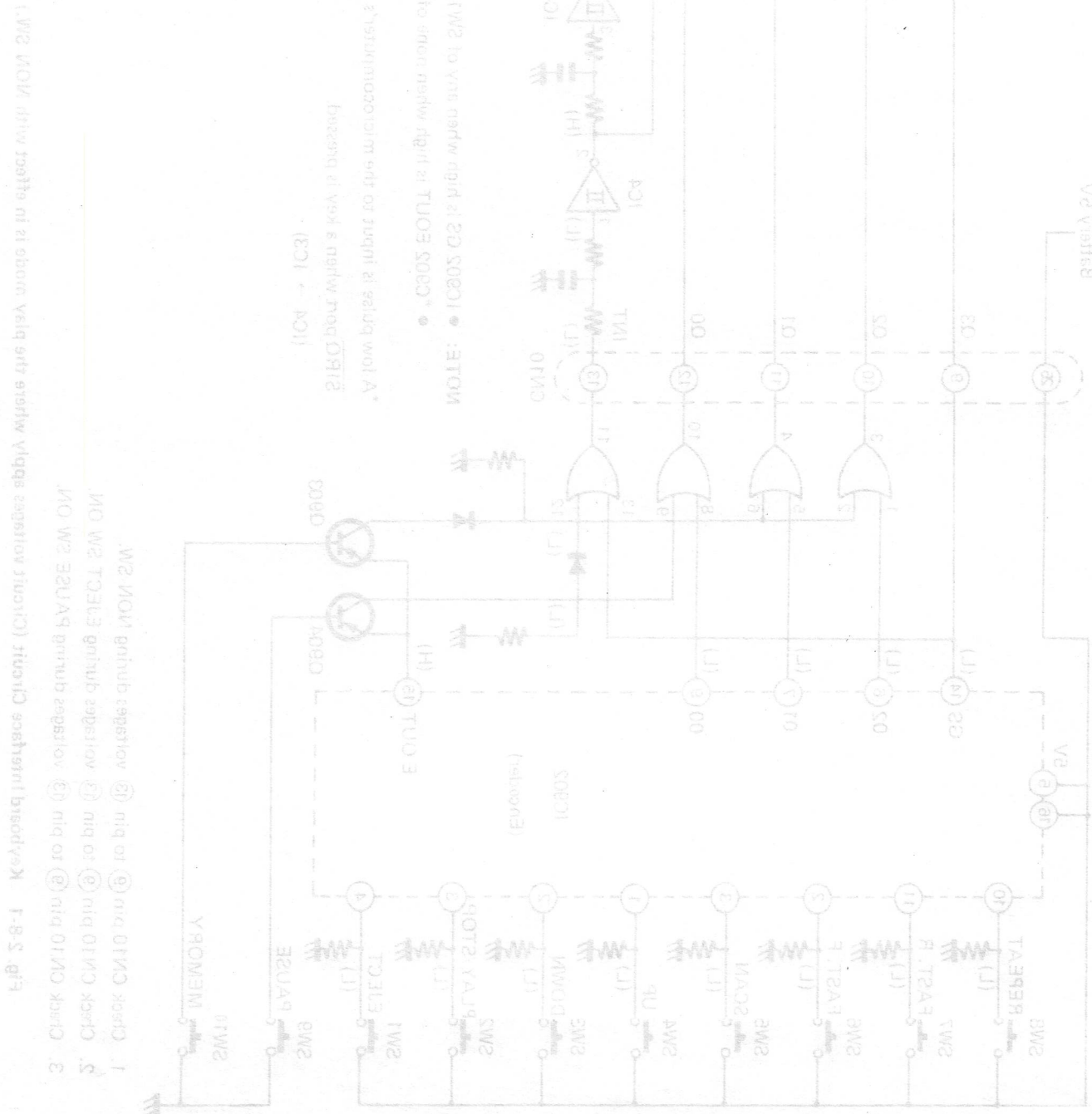
Deck Switch Input CN2 (7-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	NC	/	/	
2	SW3 input	2V or less	→	
3	Digital 5V	5V	→	0 when disk is off
4	GND	0	→	
5	SW2 input	2V or less	→	5V when disk is off
6	Accessory 5V	5V	→	
7	Door SW input	5V	→	2V or less when door is open

2-8 Checking Various Operations (Key decoder circuit)

Keyboard commands are received by the microcomputer, which sends control signals suitable to the state to initiate operate. The control signals from the microcomputer are considered nearly normal in view of the checks made so far, check the circuitry from the keyboard to the microcomputer.

See the keyboard interface circuit diagram.



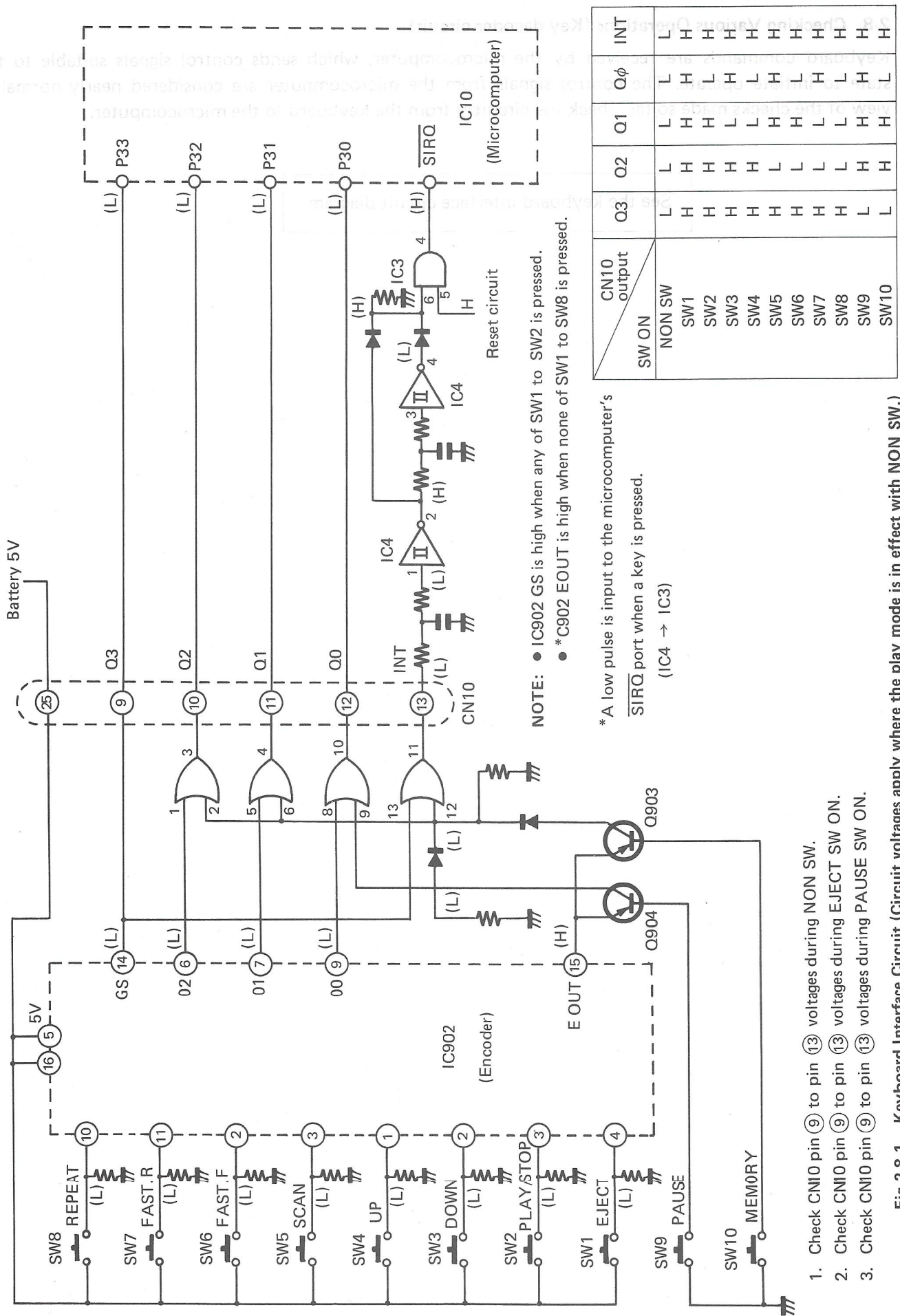


Fig.2-8-1 Keyboard Interface Circuit (Circuit voltages apply where the play mode is in effect with NON SW.)

1. Check CN10 pin ⑨ to pin ⑬ voltages during NON SW.
2. Check CN10 pin ⑨ to pin ⑬ voltages during EJECT SW ON.
3. Check CN10 pin ⑨ to pin ⑬ voltages during PAUSE SW ON.

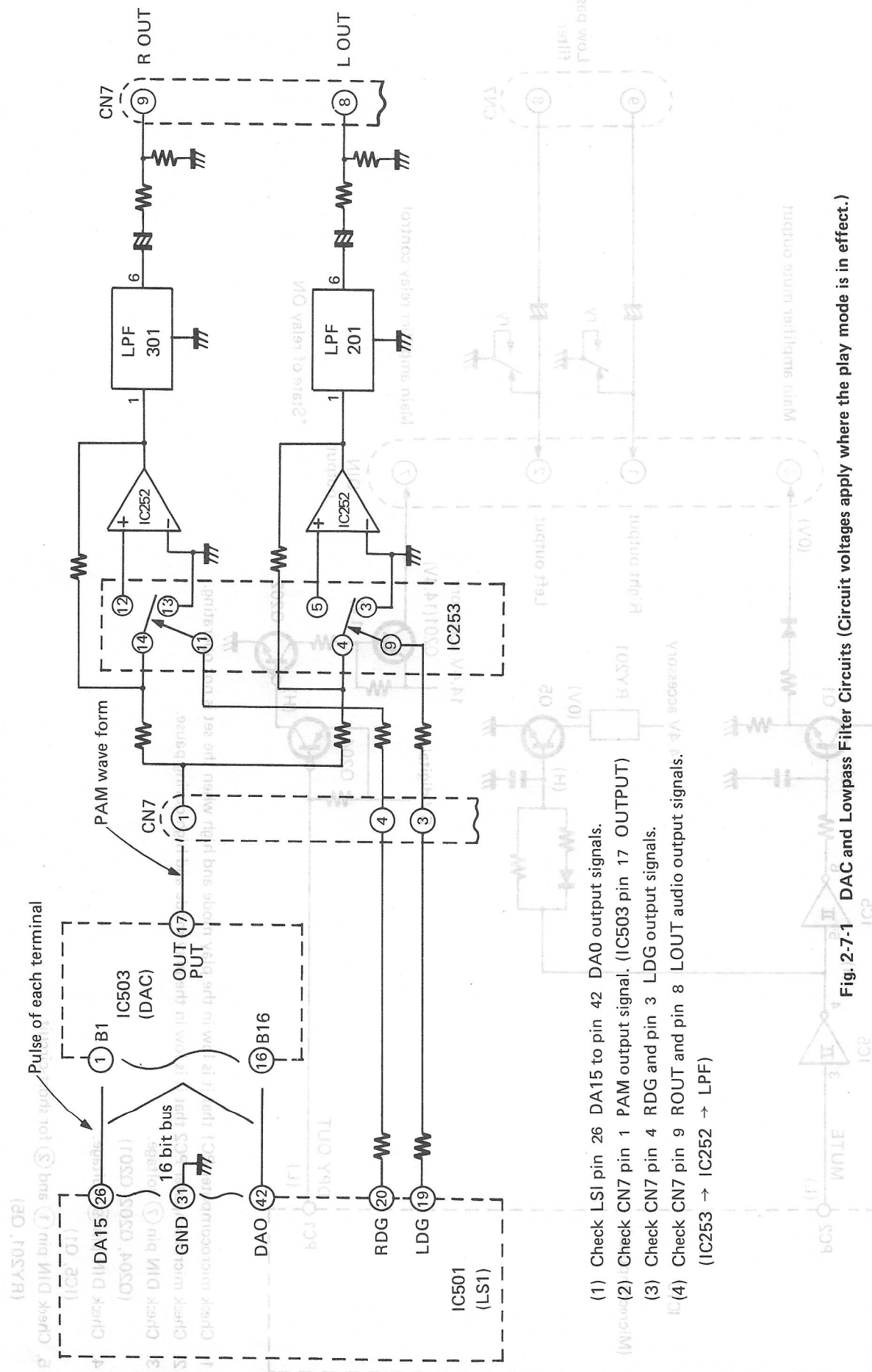
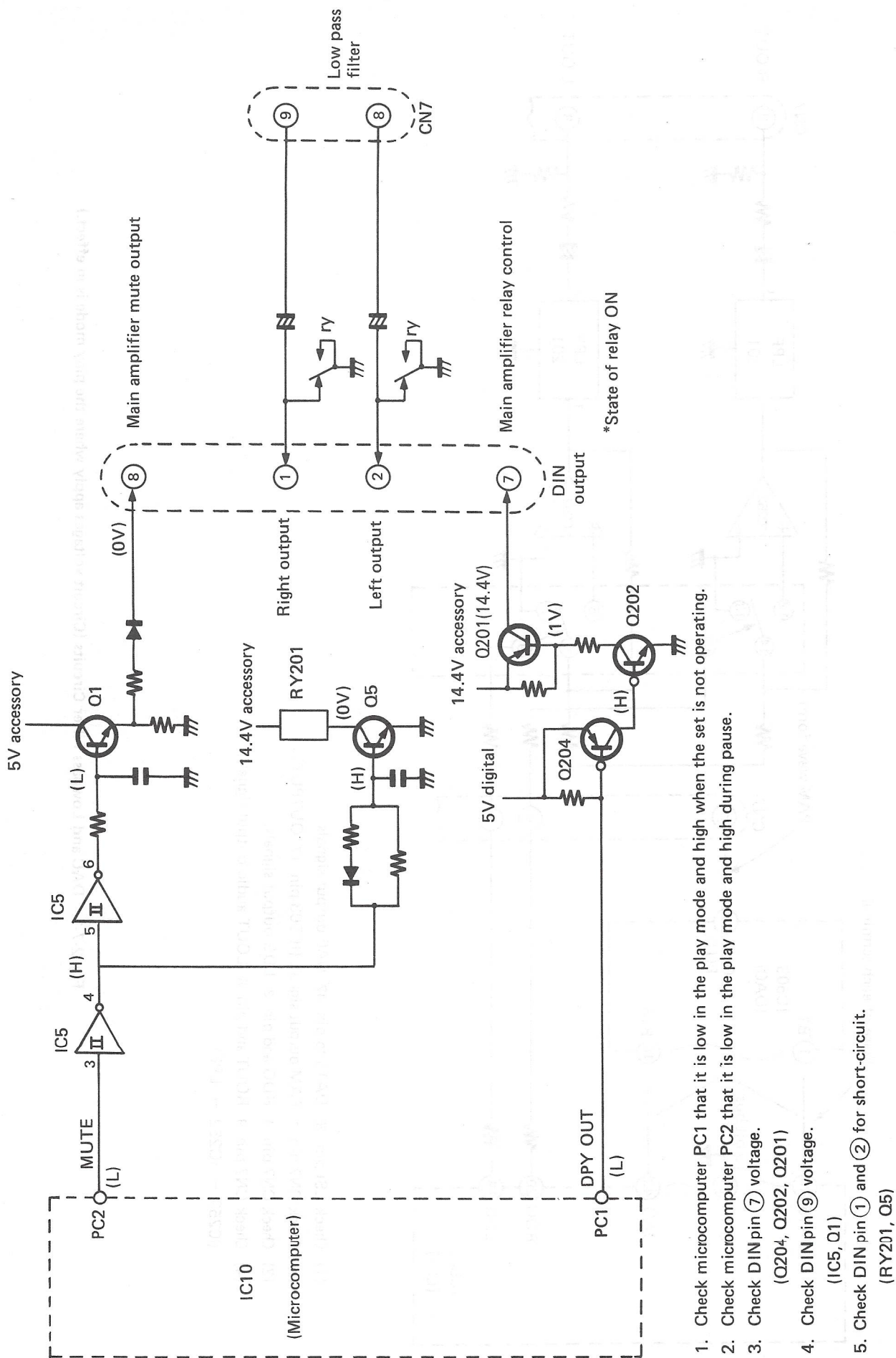


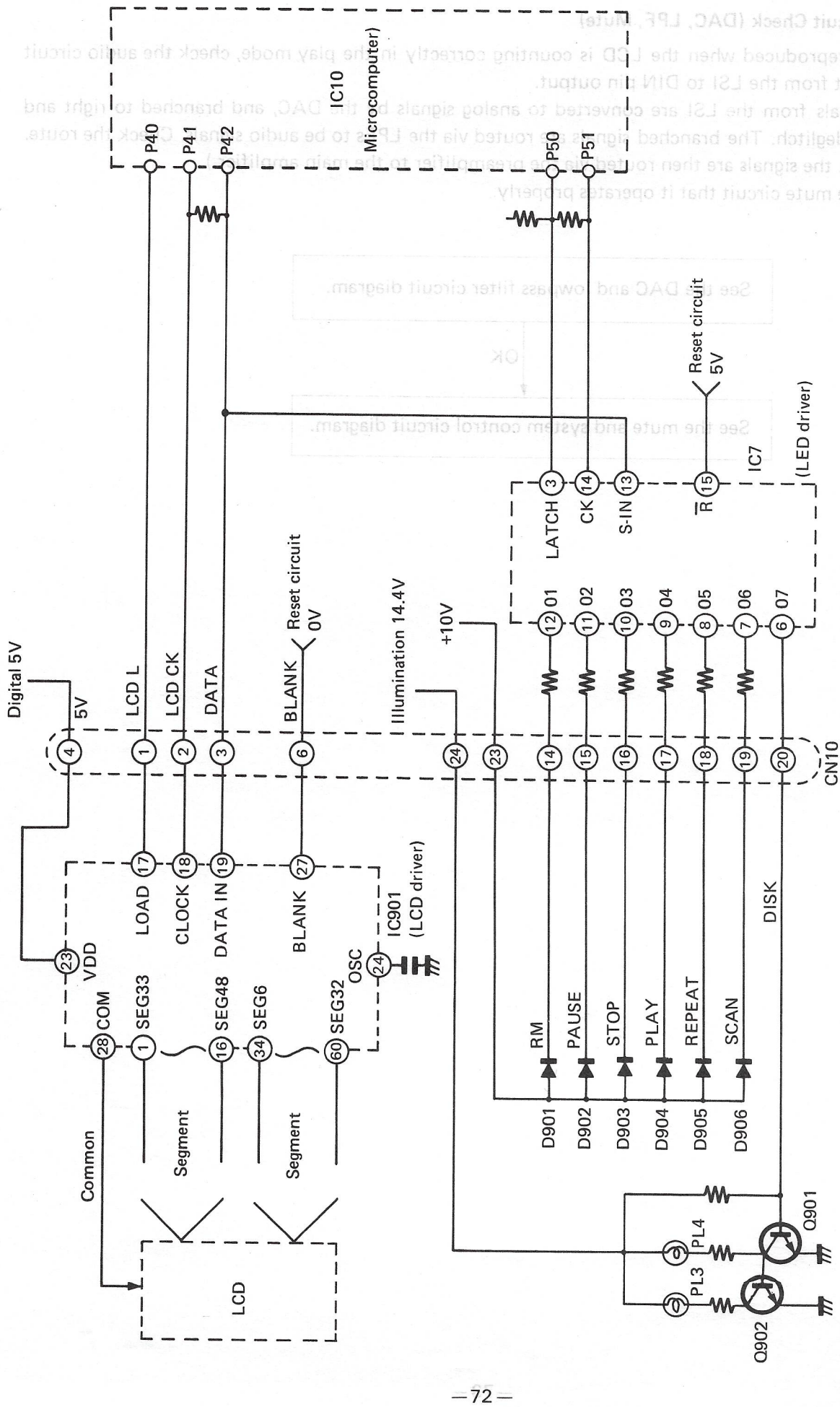
Fig. 2-7-1 DAC and Lowpass Filter Circuits (Circuit voltages apply where the play mode is in effect.)

- (1) Check LSI pin 26 DA15 to pin 42 DAO output signals.
- (2) Check CN7 pin 1 PAM output signal. (IC503 pin 17 OUTPUT)
- (3) Check CN7 pin 4 RDG and pin 3 LDG output signals.
- (4) Check CN7 pin 9 ROUT and pin 8 LOUT audio output signals.
(IC253 → IC252 → LPF)



1. Check microcomputer PC1 that it is low in the play mode and high when the set is not operating.
2. Check microcomputer PC2 that it is low in the play mode and high during pause.
3. Check DIN pin ⑦ voltage.
(Q204, Q202, Q201)
4. Check DIN pin ⑨ voltage.
(IC5, Q1)
5. Check DIN pin ① and ② for short-circuit.
(RY201, Q5)

Fig. 2-7-2 Mute and System Control Circuits (Circuit voltages apply where the play mode is in effect.)



LCD circuit check

1. Check the LCDL, LCDCK, and DATA signals from the microcomputer in the play mode.
2. Check CN10 pin 6 BLANK.
3. Check pulse waves at COM and SEG pins.
4. Check OSC and VDD.

LED circuit check

1. Check the LATCH, CK, and SIN signals from the microcomputer in the play mode.
2. Check IC7 pin 15 $\bar{R}$.
3. Check IC7 pin 9 04 that it is 0V in the play mode.
4. PL3 and PL4 light only when main amplifier illumination power is applied.

Fig. 2-6-1 LCD and LED Circuits (Circuits voltages apply where the play mode is in effect.)

2-7 Audio Circuit Check (DAC, LPF, Mute)

If no sound is reproduced when the LCD is counting correctly in the play mode, check the audio circuit and mute circuit from the LSI to DIN pin output.

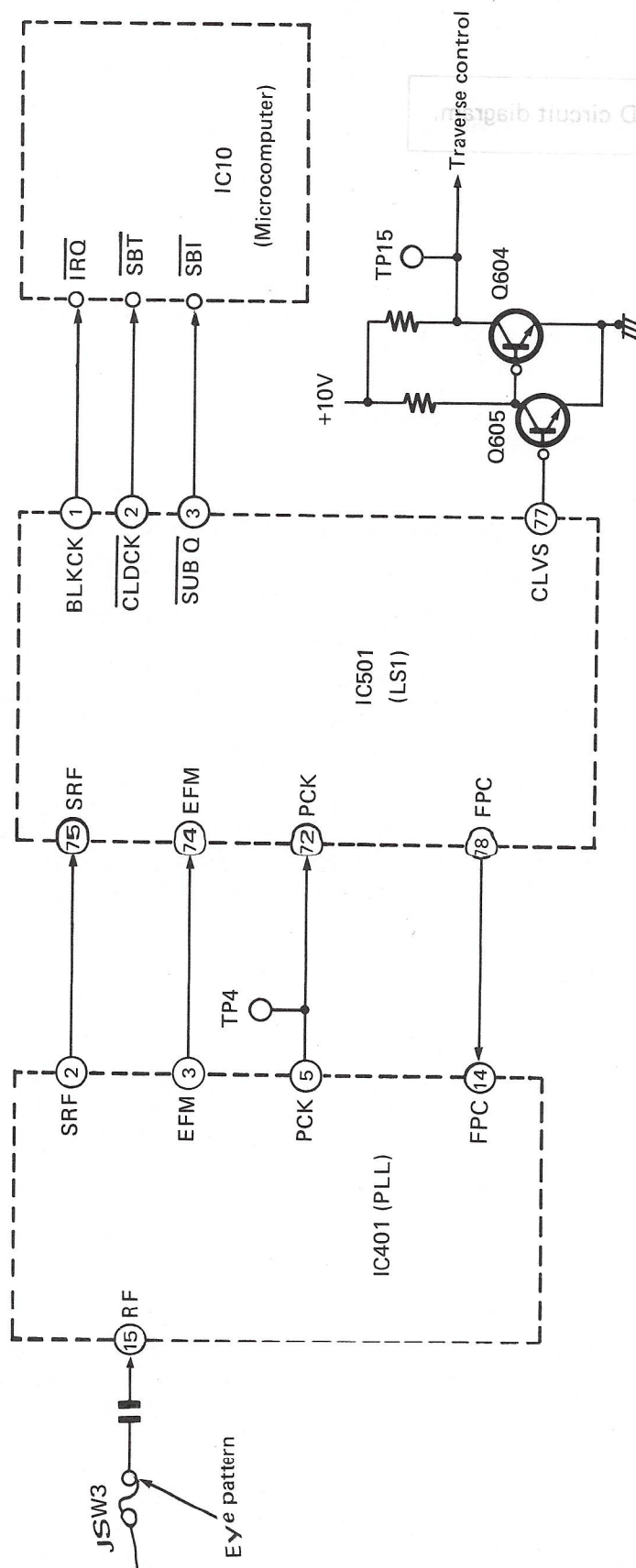
The 16-bit signals from the LSI are converted to analog signals by the DAC, and branched to right and left signals by deglitch. The branched signals are routed via the LPFs to be audio signals. Check the route. (In case of DP3, the signals are then routed via the preamplifier to the main amplifier.)

Then, check the mute circuit that it operates properly.

See the DAC and lowpass filter circuit diagram.

OK

See the mute and system control circuit diagram.



1. RF signal (JSW3) amplitude and eye pattern opening are not sufficient. Readjust the eye pattern.
2. Check PLL IC pin 3 EFM and pin 5 PCK.
3. Check LSI decision output pin 78 FPC and pin 77 CLVS.
 - ① CLVS Normal, FPC NG → PLL IC is faulty.
 - ② LLVS NG, FPC NG → Traverse deck (spindle circuit board) is faulty.
 4. Check ① BLKCK, ② CLDCK, and ③ SUBQ from LSI.

Fig. 2-5-2 Subcode Q Data Read Circuit (Circuit voltages apply where the play mode is in effect.)

2-6 LCD and LED Check (LCD driver, LED driver)

When TOC is deciphered, data is fed from the microcomputer via the drivers to select the ON mode. The LCD flashes until the play mode starts.

See the LCD and LED circuit diagram.



1) RF waveform when the focus servo is faulty.



Check points

- Microcomputer input and output ports
- Connection from IC101 pin ③⑥ FEOUT to IC102 pin 2
- Focus offset readjustment
- Focus gain readjustment

2) RF waveform when the tracking servo is faulty.



Check points

- Tracking servo circuit from IC101 pin ⑥ TEOUT to pin ⑮ HIN, pin ⑰ HOUT, CN4 pins ① and ②
- Tracking gain readjustment

3) Normal waveform

